Performance Test — Executive Summary

11m 24s	+636 MB (+24%)	4567 MB	31%	91	0
Test Duration	Memory Growth	Peak Memory	Avg CPU	Avg FPS	Console Errors

Health Check

- Memory: possible leak — 2602→3238 MB (+24%), peak 4567 MB
- FPS: smooth — avg 91, min 38
- CPU: healthy — avg 31%, peak 63%
- Console errors: none captured

Per-Phase Comparison

Segment	Mem avg	Mem peak	CPU avg	FPS avg	FPS min
Phase 1: Globe / Satellite	3274 MB	4372 MB	32%	85	38
Phase 2: Mercator / Dark	3042 MB	4567 MB	32%	97	60

Test Configuration

Run

App: https://drop-black-demo.weather.com/

Desk: None (baseline)

Phase duration: 5 min per phase

Timebar test: Off

Machine & display

OS: macOS 25.6.0 (arm64) · CPU: Apple M3 Pro (12 cores) · RAM: 36 GB · Host: TWC-M2HGDHWL4V

Monitor: 1718×960 CSS px @ 1x · 30-bit color · 121 Hz (max ~121 FPS)

Browser: Chromium 151.0.7922.34 · Test viewport: 1718×960

Phases

- Phase 1: Globe / Satellite
- Phase 4: Mercator / Dark

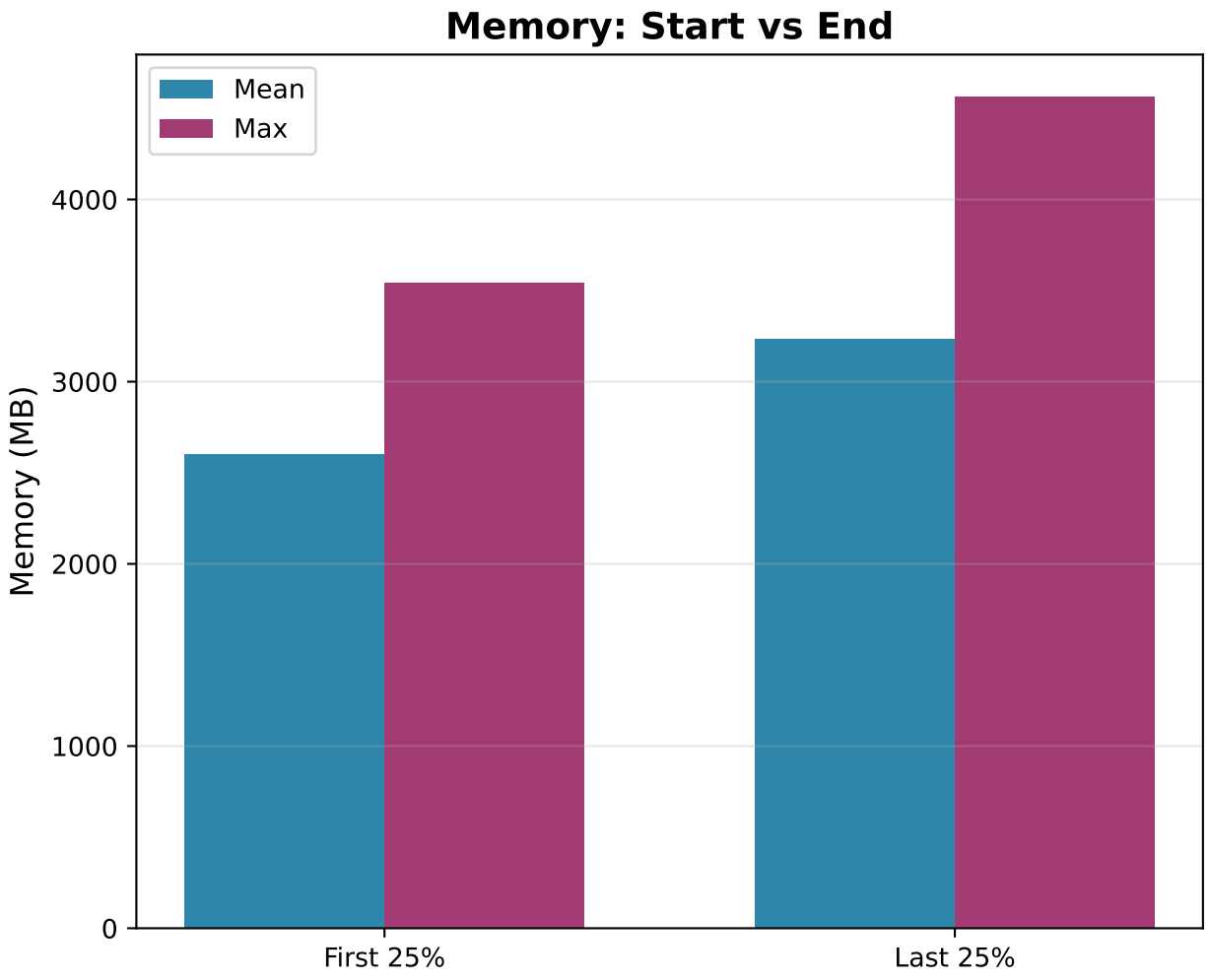
Layer (Layers tab)

Radar

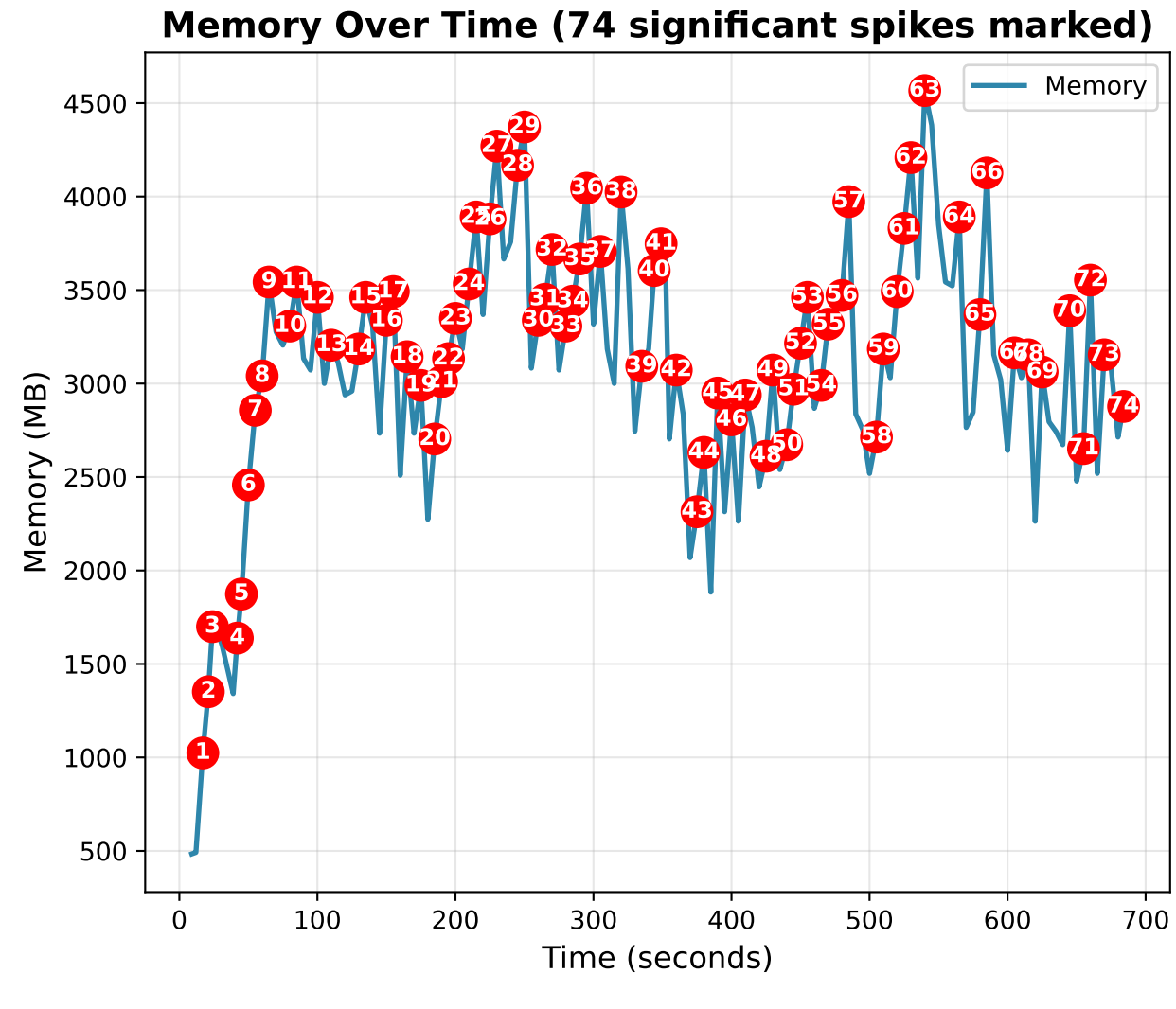
Overlays ON (4)

Lightning, Airports, Cman Buoys, Ship Reports

Memory Performance Analysis

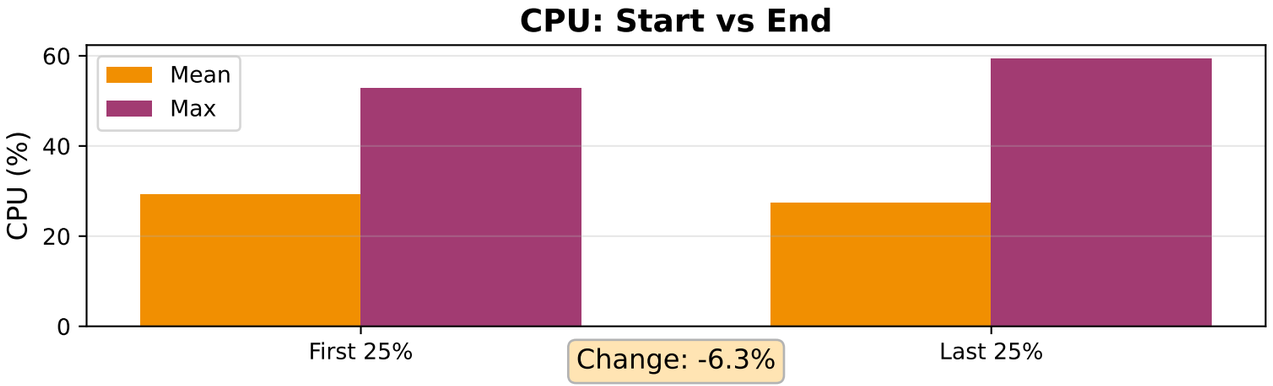


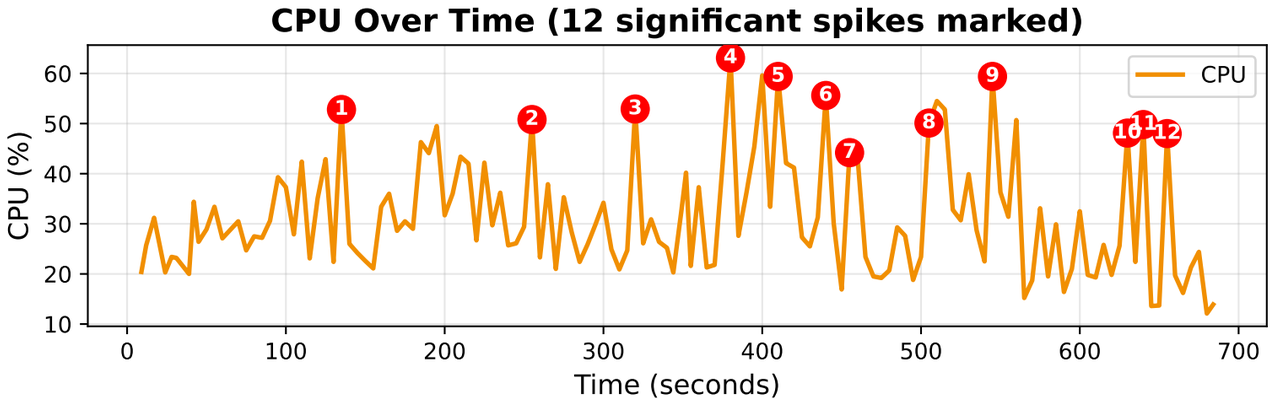
Change: +24.4%




```
Memory Spike Events:
  1. 17s: Radar enabled (1024 MB)
  2. 21s: Lightning enabled (1352 MB)
  3. 24s: Airports enabled (1700 MB)
  4. 42s: Set to Globe/Satellite (1638 MB)
  5. 45s: Panning and zooming (1874 MB)
  6. 50s: Panning and zooming (2458 MB)
  7. 55s: Panning and zooming (2857 MB)
  8. 60s: Panning and zooming (3041 MB)
  9. 65s: Panning and zooming (3543 MB)
  10. 80s: Pan only (L/R) (3308 MB)
  11. 85s: Pan only (L/R) (3543 MB)
  12. 100s: Zoom only (in/out) (3461 MB)
  13. 110s: Panning and zooming (3205 MB)
  14. 130s: Panning and zooming (3185 MB)
  15. 135s: Panning and zooming (3461 MB)
  16. 150s: Pan only (L/R) (3338 MB)
  17. 155s: Pan only (L/R) (3492 MB)
  18. 165s: Zoom only (in/out) (3144 MB)
  19. 175s: Panning and zooming (2990 MB)
  20. 185s: Panning and zooming (2703 MB)
  21. 190s: Panning and zooming (3011 MB)
  22. 195s: Panning and zooming (3133 MB)
  23. 200s: Panning and zooming (3348 MB)
  24. 210s: Panning and zooming (3533 MB)
  25. 215s: Pan only (L/R) (3891 MB)
  26. 225s: Pan only (L/R) (3881 MB)
  27. 230s: Zoom only (in/out) (4270 MB)
  28. 245s: Panning and zooming (4168 MB)
  29. 250s: Panning and zooming (4372 MB)
  30. 260s: Panning and zooming (3338 MB)
  31. 265s: Panning and zooming (3451 MB)
  32. 270s: Panning and zooming (3717 MB)
  33. 280s: Pan only (L/R) (3308 MB)
  34. 285s: Pan only (L/R) (3441 MB)
  35. 290s: Pan only (L/R) (3666 MB)
  36. 295s: Zoom only (in/out) (4045 MB)
  37. 305s: Zoom only (in/out) (3707 MB)
  38. 320s: Panning and zooming (4024 MB)
  39. 335s: Panning and zooming (3092 MB)
  40. 344s: Changing map mode (3604 MB)
  41. 349s: Changing map mode (3748 MB)
  42. 360s: Panning and zooming (3072 MB)
  43. 375s: Panning and zooming (2314 MB)
  44. 380s: Pan only (L/R) (2632 MB)
  45. 390s: Pan only (L/R) (2949 MB)
  46. 400s: Zoom only (in/out) (2806 MB)
  47. 410s: Panning and zooming (2939 MB)
  48. 425s: Panning and zooming (2611 MB)
  49. 430s: Panning and zooming (3072 MB)
  50. 440s: Panning and zooming (2673 MB)
  51. 445s: Panning and zooming (2970 MB)
  52. 450s: Pan only (L/R) (3215 MB)
  53. 455s: Pan only (L/R) (3461 MB)
  54. 465s: Zoom only (in/out) (2990 MB)
  55. 470s: Zoom only (in/out) (3318 MB)
  56. 480s: Panning and zooming (3471 MB)
  57. 485s: Panning and zooming (3973 MB)
  58. 505s: Panning and zooming (2714 MB)
  59. 510s: Panning and zooming (3185 MB)
  60. 520s: Pan only (L/R) (3492 MB)
  61. 525s: Pan only (L/R) (3830 MB)
  62. 530s: Zoom only (in/out) (4209 MB)
  63. 540s: Zoom only (in/out) (4567 MB)
  64. 565s: Panning and zooming (3891 MB)
  65. 580s: Pan only (L/R) (3369 MB)
  66. 585s: Pan only (L/R) (4127 MB)
  67. 605s: Zoom only (in/out) (3164 MB)
  68. 615s: Panning and zooming (3154 MB)
  69. 625s: Panning and zooming (3062 MB)
  70. 645s: Panning and zooming (3389 MB)
  71. 655s: Pan only (L/R) (2652 MB)
  72. 660s: Pan only (L/R) (3553 MB)
  73. 670s: Zoom only (in/out) (3154 MB)
  74. 684s: Test complete (2877 MB)
```

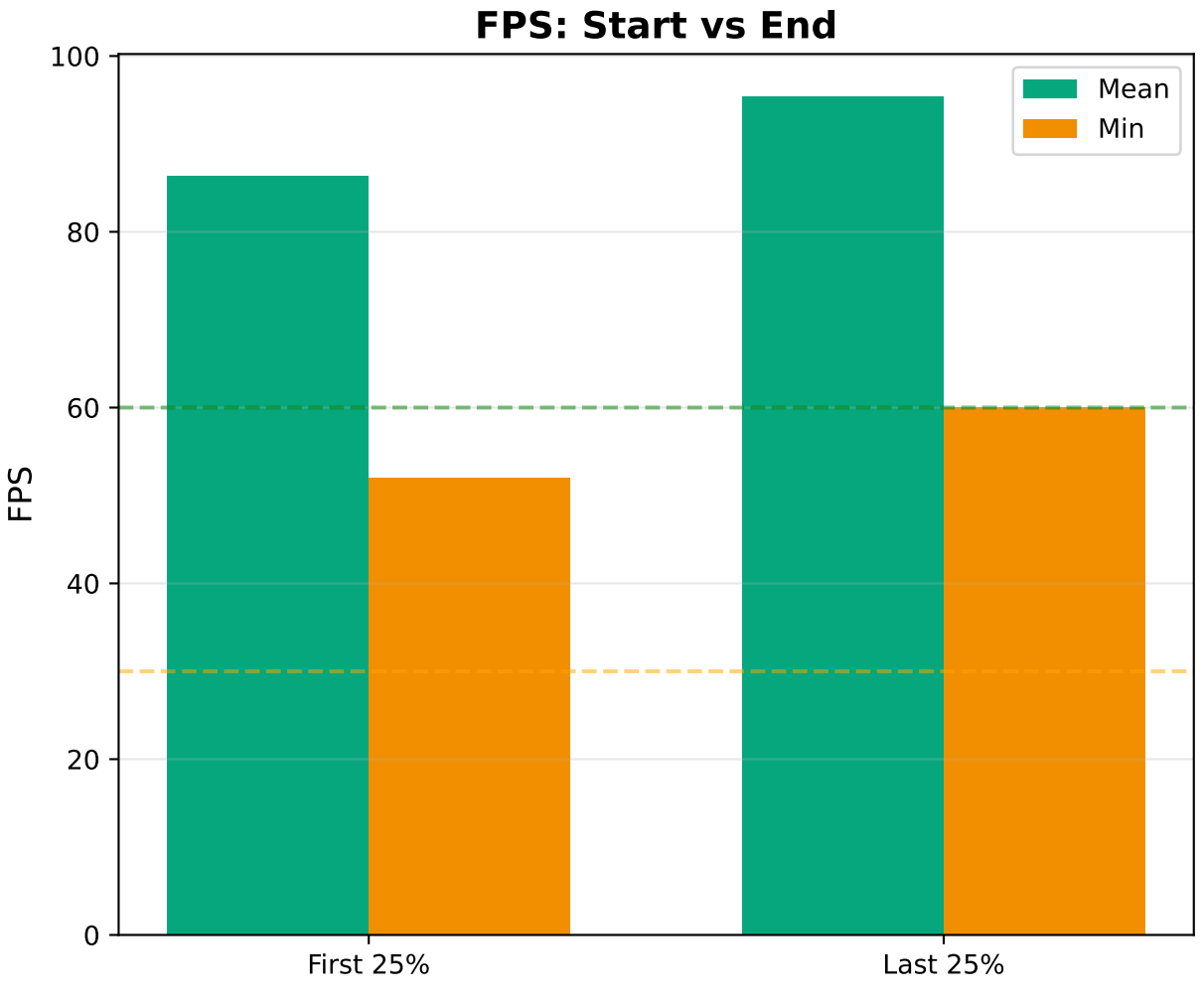
CPU Performance Analysis



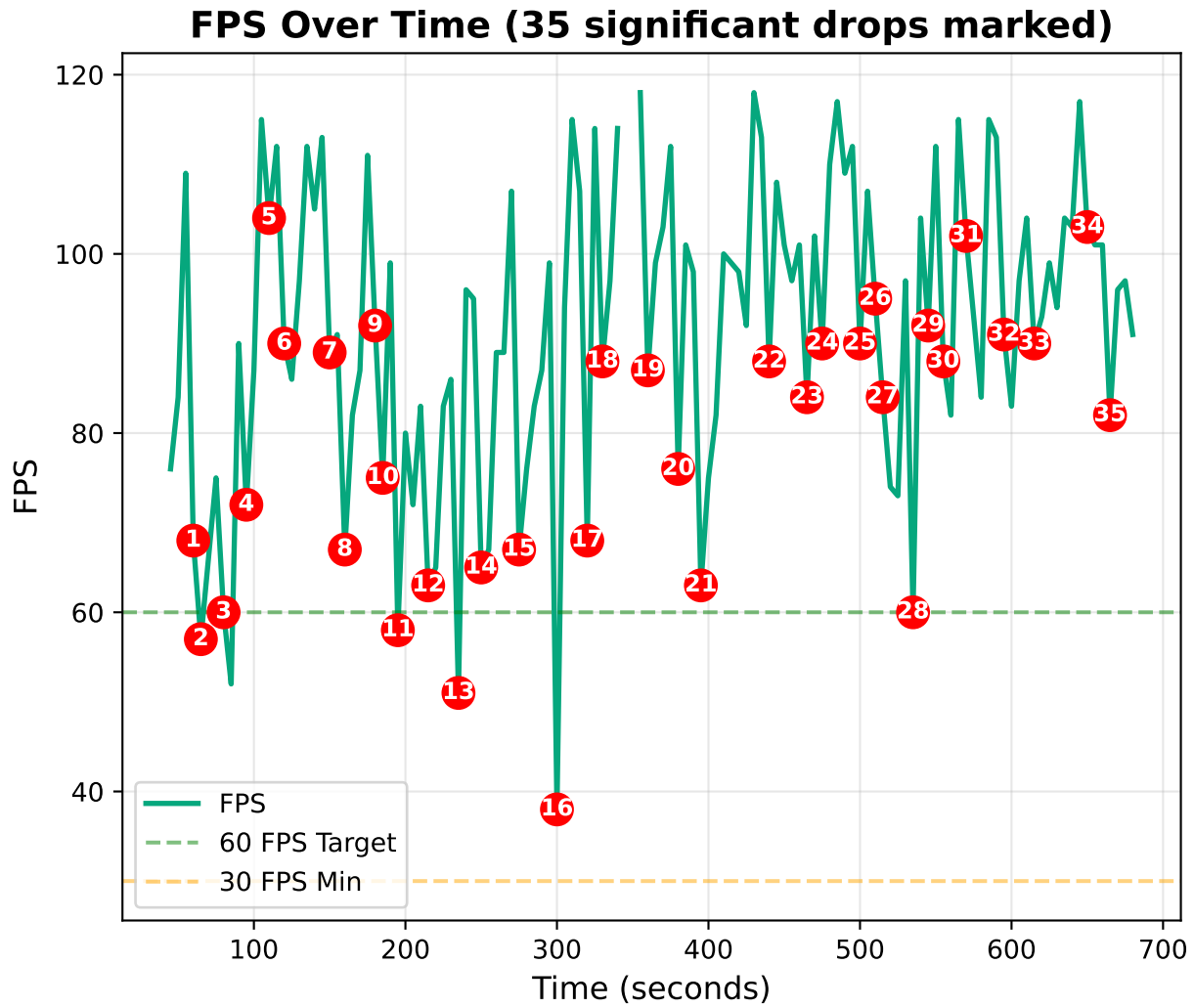



```
CPU Spike Events:
  1. 135s: Panning and zooming (53%)
  2. 255s: Panning and zooming (51%)
  3. 320s: Panning and zooming (53%)
  4. 380s: Pan only (L/R) (63%)
  5. 410s: Panning and zooming (59%)
  6. 440s: Panning and zooming (56%)
  7. 455s: Pan only (L/R) (44%)
  8. 505s: Panning and zooming (50%)
  9. 545s: Panning and zooming (59%)
  10. 630s: Panning and zooming (48%)
  11. 640s: Panning and zooming (50%)
  12. 655s: Pan only (L/R) (48%)
```

FPS Performance Analysis

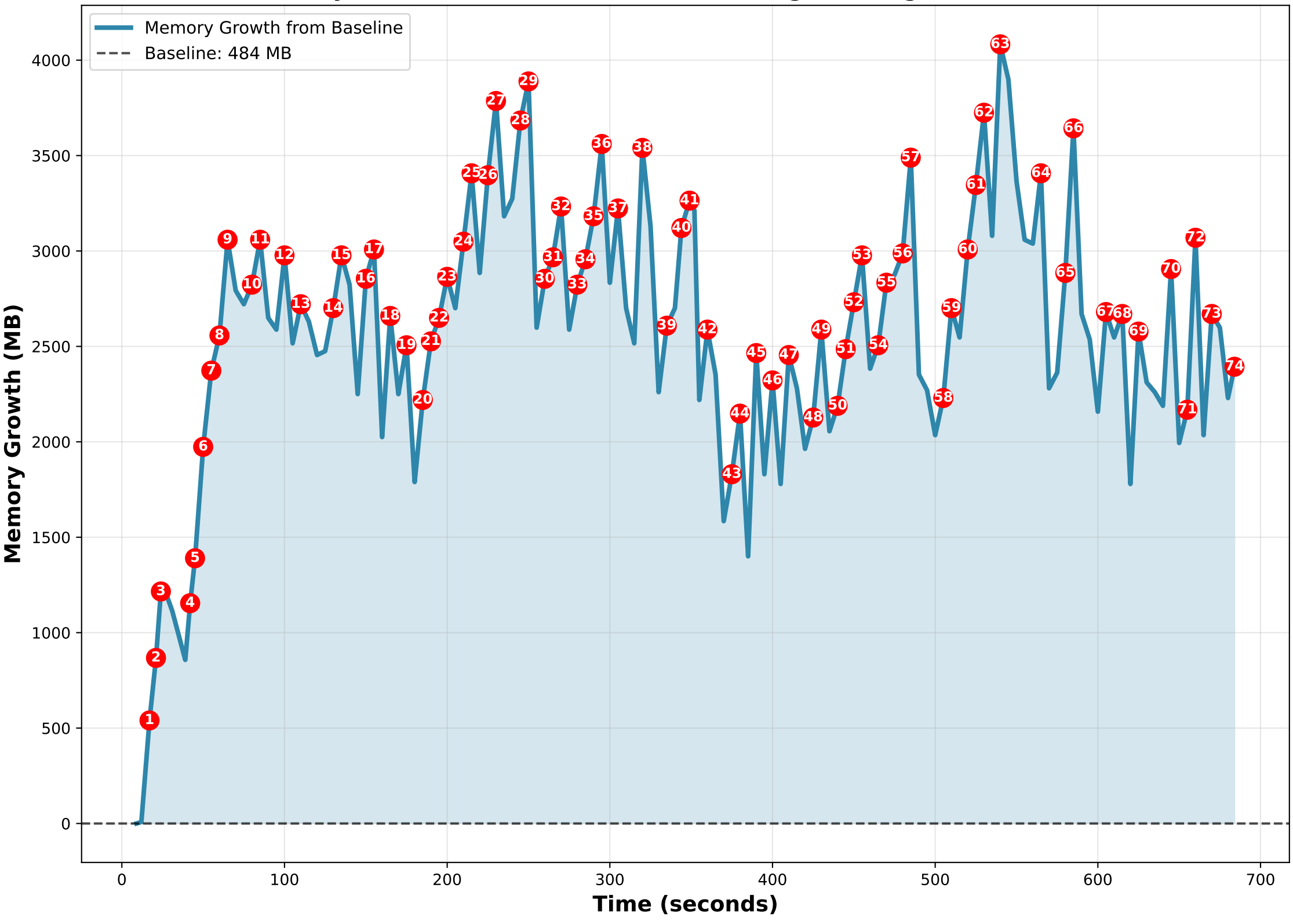


Change: +10.5%




```
FPS Drop Events:
  1. 60s: Panning and zooming (68 FPS)
  2. 65s: Panning and zooming (57 FPS)
  3. 80s: Pan only (L/R) (60 FPS)
  4. 95s: Zoom only (in/out) (72 FPS)
  5. 110s: Panning and zooming (104 FPS)
  6. 120s: Panning and zooming (90 FPS)
  7. 150s: Pan only (L/R) (89 FPS)
  8. 160s: Zoom only (in/out) (67 FPS)
  9. 180s: Panning and zooming (92 FPS)
  10. 185s: Panning and zooming (75 FPS)
  11. 195s: Panning and zooming (58 FPS)
  12. 215s: Pan only (L/R) (63 FPS)
  13. 235s: Zoom only (in/out) (51 FPS)
  14. 250s: Panning and zooming (65 FPS)
  15. 275s: Panning and zooming (67 FPS)
  16. 300s: Zoom only (in/out) (38 FPS)
  17. 320s: Panning and zooming (68 FPS)
  18. 330s: Panning and zooming (88 FPS)
  19. 360s: Panning and zooming (87 FPS)
  20. 380s: Pan only (L/R) (76 FPS)
  21. 395s: Zoom only (in/out) (63 FPS)
  22. 440s: Panning and zooming (88 FPS)
  23. 465s: Zoom only (in/out) (84 FPS)
  24. 475s: Zoom only (in/out) (90 FPS)
  25. 500s: Panning and zooming (90 FPS)
  26. 510s: Panning and zooming (95 FPS)
  27. 515s: Pan only (L/R) (84 FPS)
  28. 535s: Zoom only (in/out) (60 FPS)
  29. 545s: Panning and zooming (92 FPS)
  30. 555s: Panning and zooming (88 FPS)
  31. 570s: Panning and zooming (102 FPS)
  32. 595s: Zoom only (in/out) (91 FPS)
  33. 615s: Panning and zooming (90 FPS)
  34. 650s: Pan only (L/R) (103 FPS)
  35. 665s: Zoom only (in/out) (82 FPS)
```

Cumulative Memory Growth Analysis



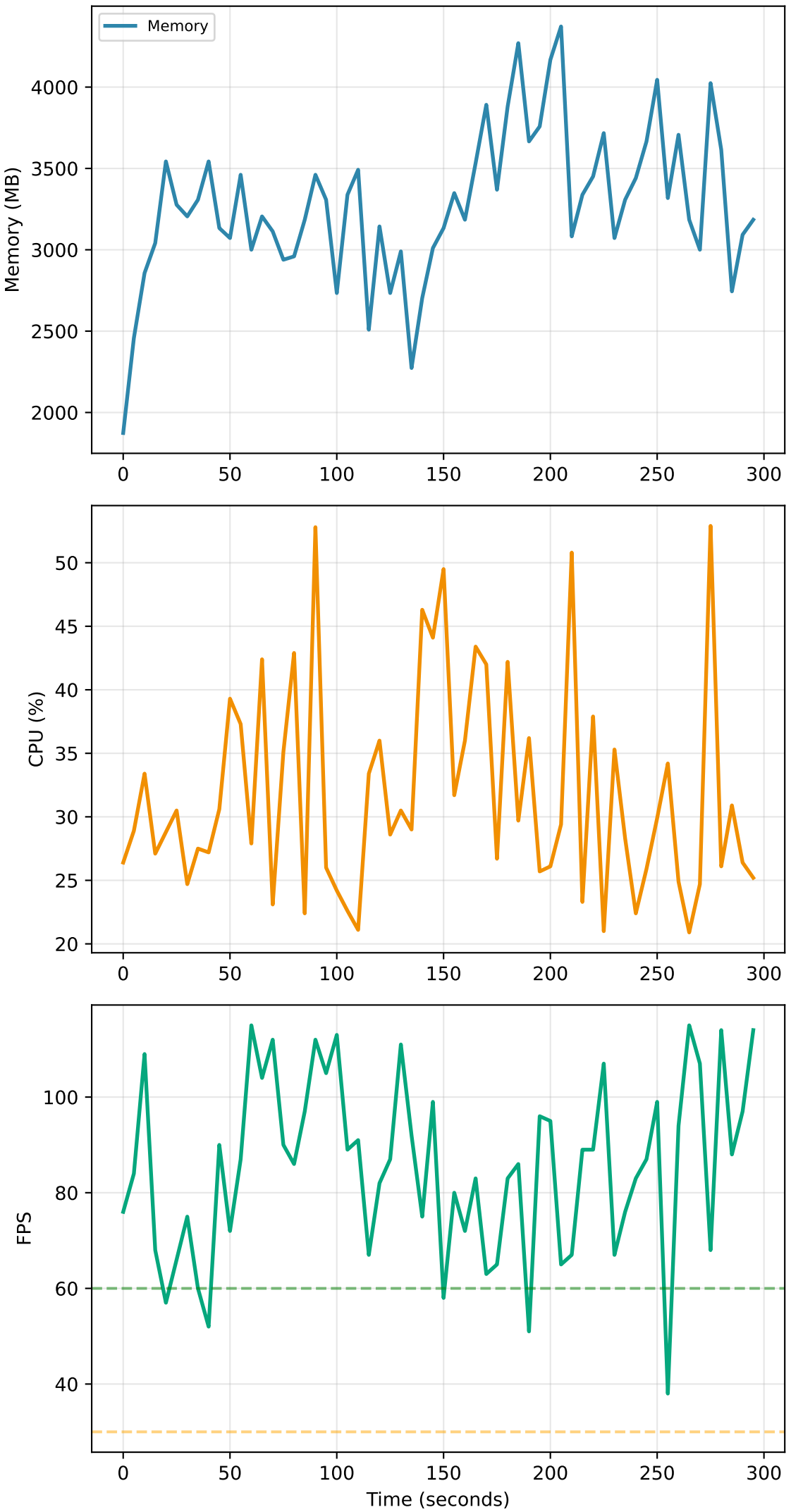
Significant Growth Events:
1. 17s: Radar enabled (+532 MB)
2. 21s: Lightning enabled (+328 MB)
3. 24s: Airports enabled (+348 MB)
4. 42s: Set to Globe/Satellite (+297 MB)
5. 45s: Panning and zooming (+236 MB)
6. 50s: Panning and zooming (+584 MB)
7. 55s: Panning and zooming (+399 MB)
8. 60s: Panning and zooming (+184 MB)
9. 65s: Panning and zooming (+502 MB)
10. 80s: Pan only (L/R) (+102 MB)
11. 85s: Pan only (L/R) (+236 MB)
12. 100s: Zoom only (in/out) (+389 MB)
13. 110s: Panning and zooming (+205 MB)
14. 130s: Panning and zooming (+225 MB)
15. 135s: Panning and zooming (+276 MB)
16. 150s: Pan only (L/R) (+604 MB)
17. 155s: Pan only (L/R) (+154 MB)
18. 165s: Zoom only (in/out) (+635 MB)
19. 175s: Panning and zooming (+256 MB)
20. 185s: Panning and zooming (+430 MB)
21. 190s: Panning and zooming (+307 MB)
22. 195s: Panning and zooming (+123 MB)
23. 200s: Panning and zooming (+215 MB)
24. 210s: Panning and zooming (+348 MB)
25. 215s: Pan only (L/R) (+358 MB)
26. 225s: Pan only (L/R) (+512 MB)
27. 230s: Zoom only (in/out) (+389 MB)
28. 245s: Panning and zooming (+410 MB)
29. 250s: Panning and zooming (+205 MB)
30. 260s: Panning and zooming (+256 MB)
31. 265s: Panning and zooming (+113 MB)
32. 270s: Panning and zooming (+266 MB)
33. 280s: Pan only (L/R) (+236 MB)
34. 285s: Pan only (L/R) (+133 MB)
35. 290s: Pan only (L/R) (+225 MB)
36. 295s: Zoom only (in/out) (+379 MB)
37. 305s: Zoom only (in/out) (+389 MB)
38. 320s: Panning and zooming (+1024 MB)
39. 335s: Panning and zooming (+348 MB)
40. 344s: Changing map mode (+420 MB)
41. 349s: Changing map mode (+143 MB)
42. 360s: Panning and zooming (+369 MB)
43. 375s: Panning and zooming (+246 MB)
44. 380s: Pan only (L/R) (+317 MB)
45. 390s: Pan only (L/R) (+1065 MB)
46. 400s: Zoom only (in/out) (+492 MB)
47. 410s: Panning and zooming (+676 MB)
48. 425s: Panning and zooming (+164 MB)
49. 430s: Panning and zooming (+461 MB)
50. 440s: Panning and zooming (+133 MB)
51. 445s: Panning and zooming (+297 MB)
52. 450s: Pan only (L/R) (+246 MB)
53. 455s: Pan only (L/R) (+246 MB)
54. 465s: Zoom only (in/out) (+123 MB)
55. 470s: Zoom only (in/out) (+328 MB)
56. 480s: Panning and zooming (+113 MB)
57. 485s: Panning and zooming (+502 MB)
58. 505s: Panning and zooming (+195 MB)
59. 510s: Panning and zooming (+471 MB)
60. 520s: Pan only (L/R) (+461 MB)
61. 525s: Pan only (L/R) (+338 MB)
62. 530s: Zoom only (in/out) (+379 MB)
63. 540s: Zoom only (in/out) (+1004 MB)
64. 565s: Panning and zooming (+369 MB)
65. 580s: Pan only (L/R) (+522 MB)
66. 585s: Pan only (L/R) (+758 MB)
67. 605s: Zoom only (in/out) (+522 MB)
68. 615s: Panning and zooming (+123 MB)
69. 625s: Panning and zooming (+799 MB)
70. 645s: Panning and zooming (+717 MB)
71. 655s: Pan only (L/R) (+174 MB)
72. 660s: Pan only (L/R) (+901 MB)
73. 670s: Zoom only (in/out) (+635 MB)
74. 684s: Test complete (+164 MB)

MEMORY GROWTH SUMMARY

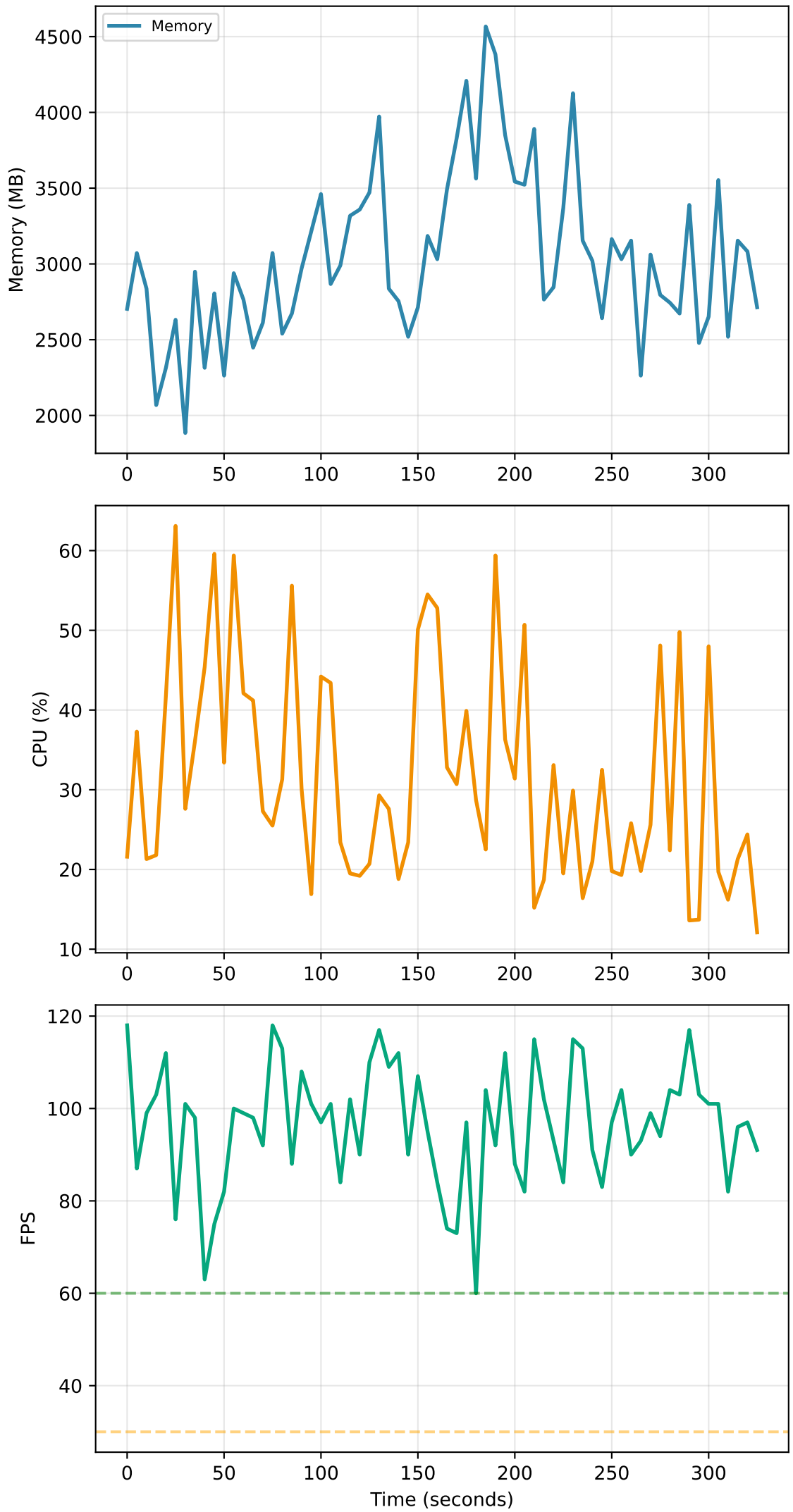
Starting Memory:	484.0 MB
Ending Memory:	2877.4 MB
Total Growth:	+2393.4 MB
Percent Increase:	+494.5%
Test Duration:	684 seconds

Phase-by-Phase Performance Comparison

Phase 1: Globe / Satellite

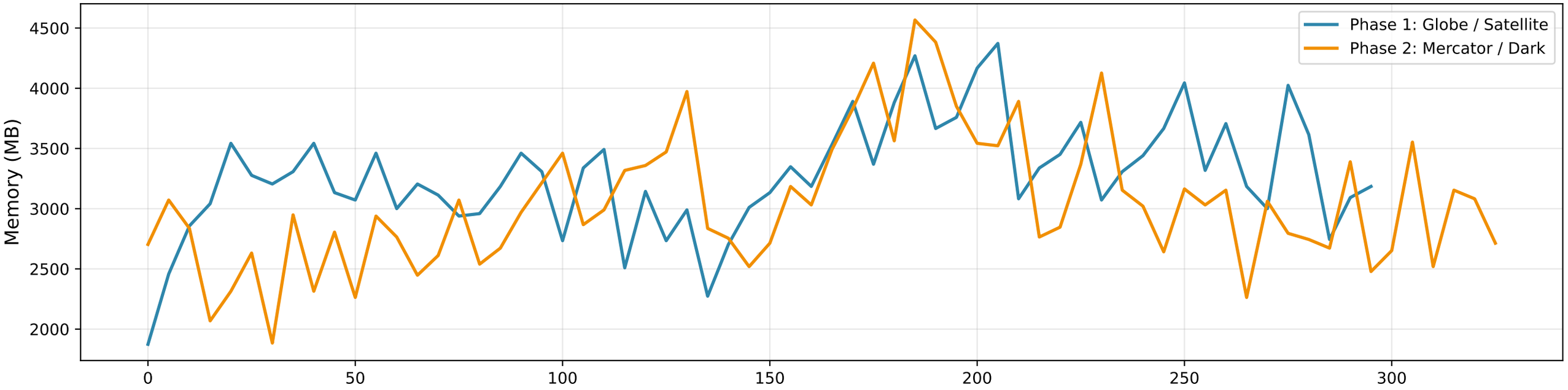


Phase 2: Mercator / Dark

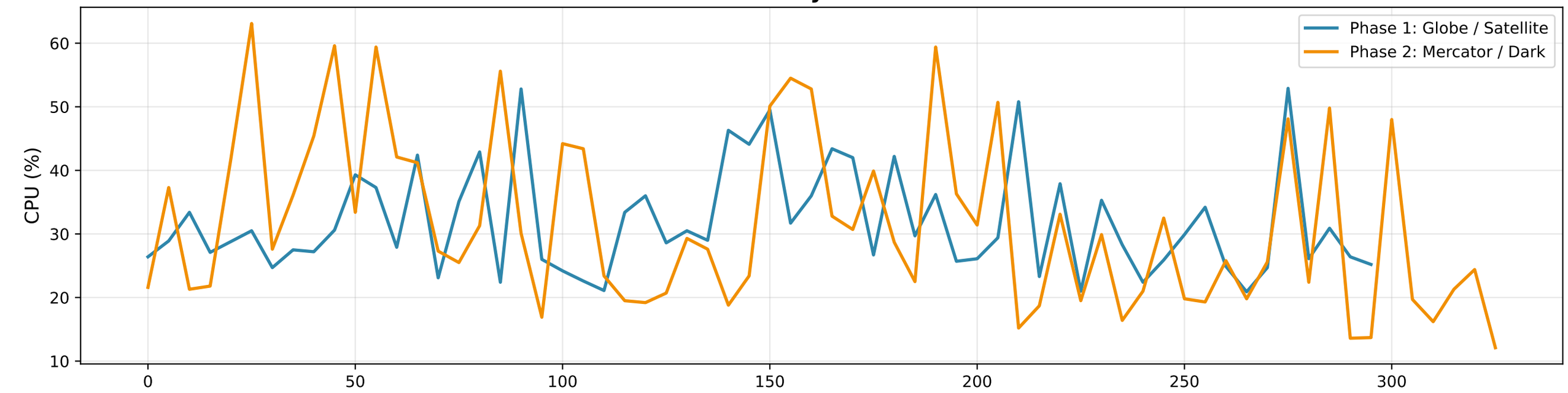


Phase Overlay Comparison

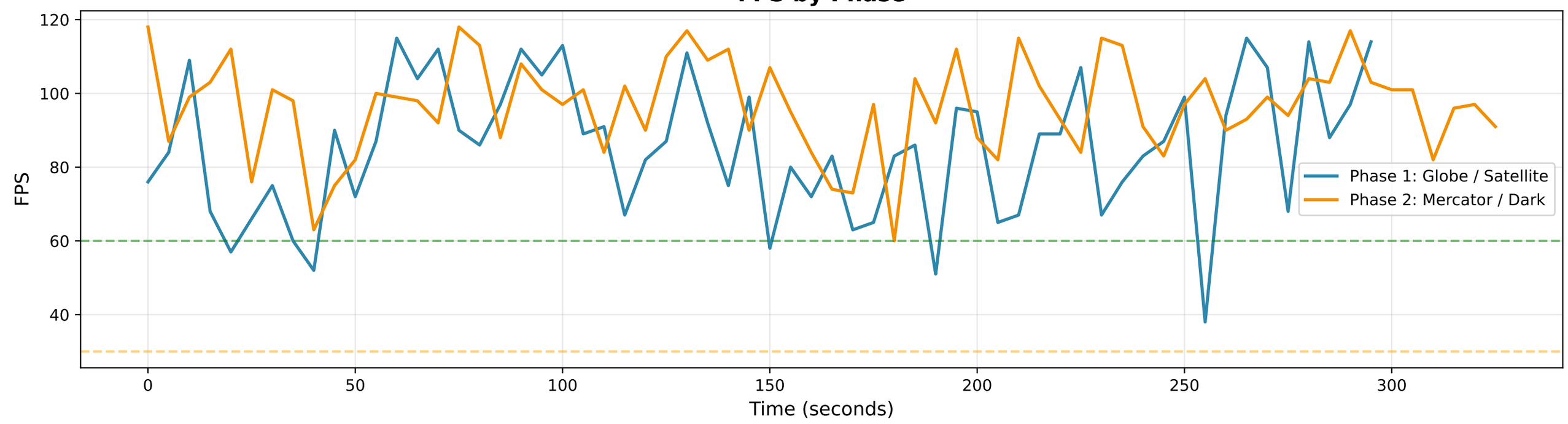
Memory by Phase



CPU by Phase

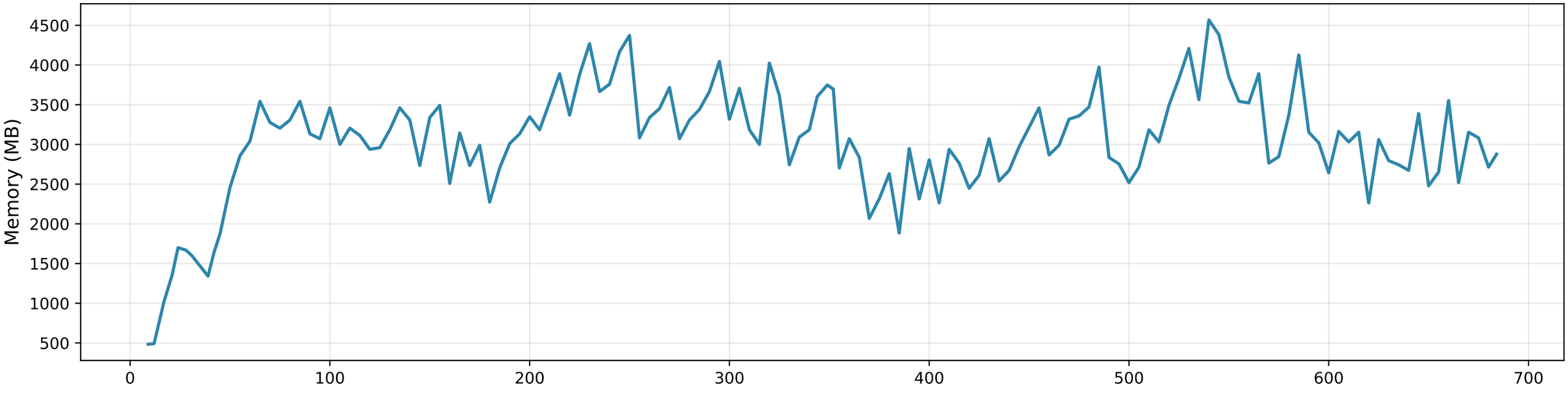


FPS by Phase

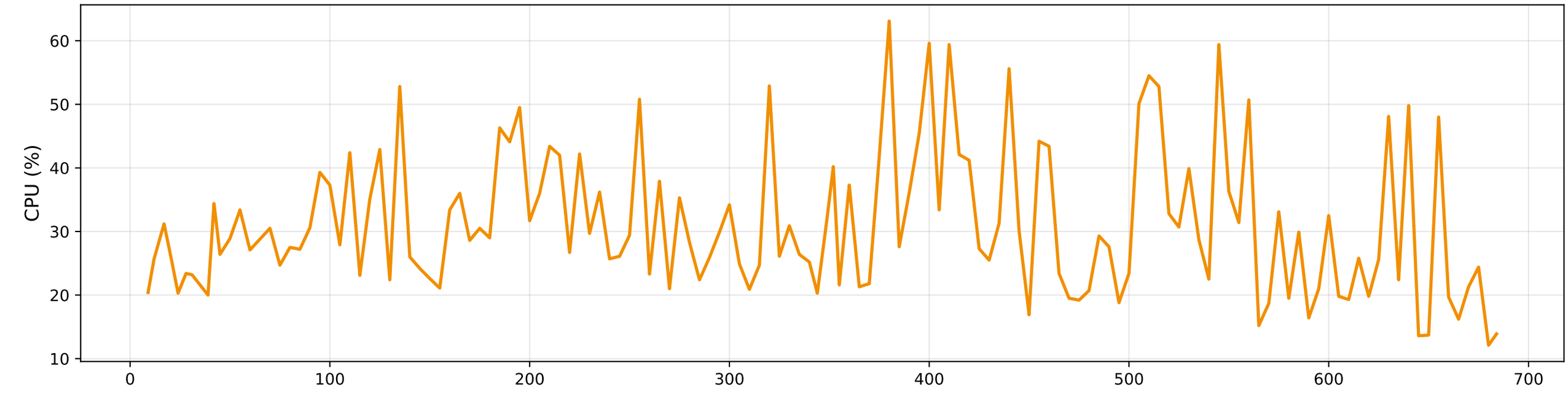


Combined Performance Timeline

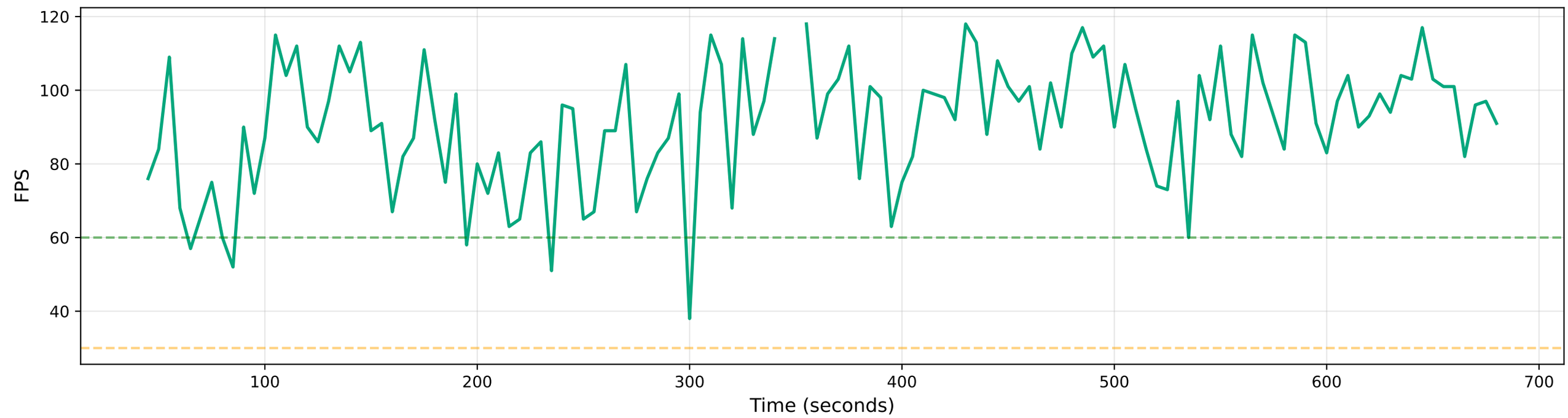
Memory Over Time

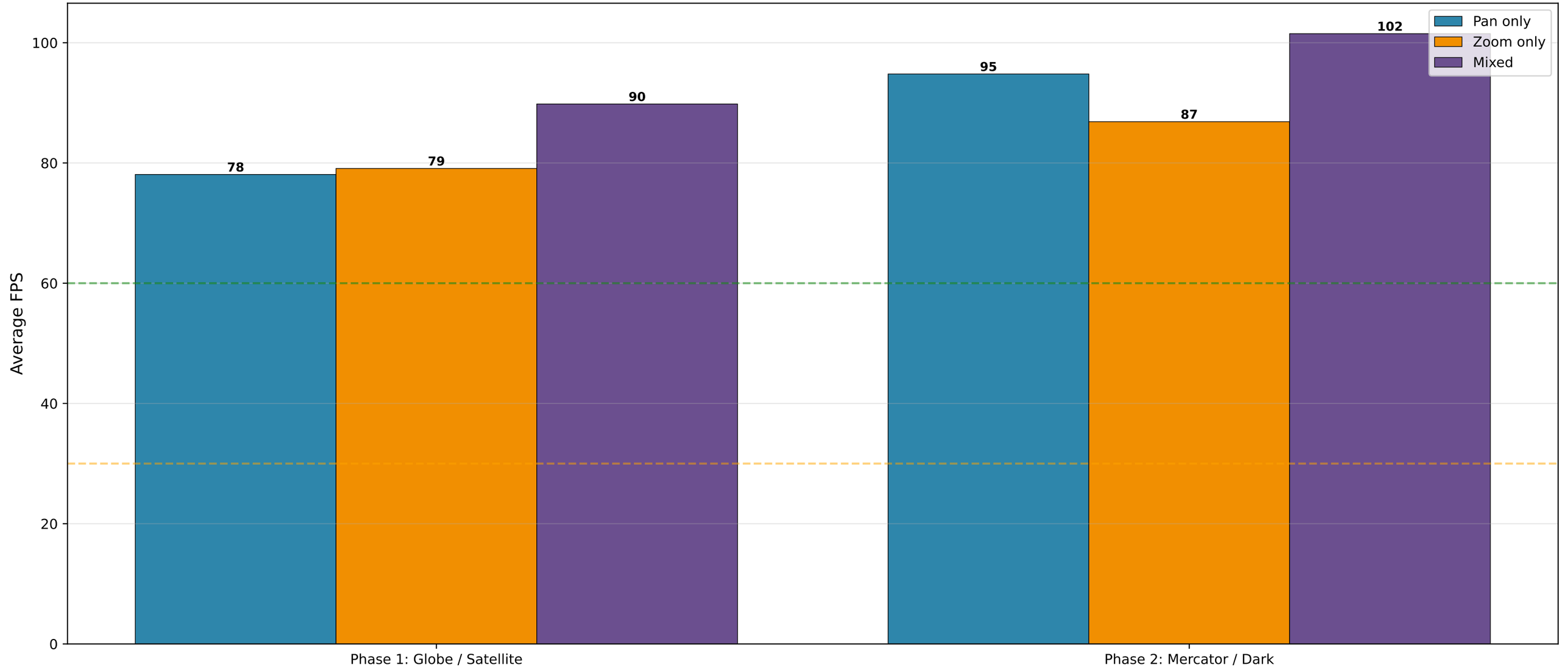


CPU Over Time



FPS Over Time





Phase	Pan only	Zoom only	Mixed
Phase 1: Globe / Satellite	78	79	90
Phase 2: Mercator / Dark	95	87	102

Console Errors by Frequency

No console errors captured during this test